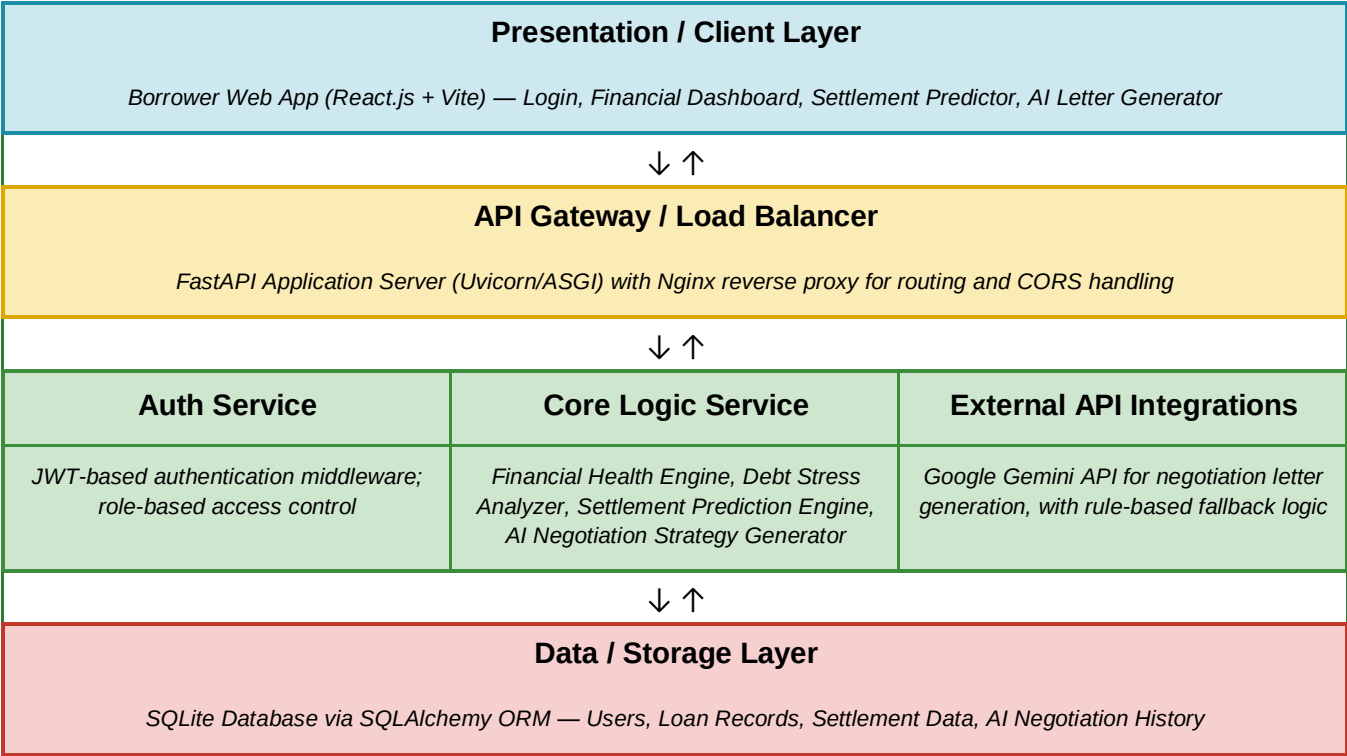


Solution Architecture

Date	12 July 2026
Team ID	6a2fbf036b1ed0bd8fa789aa
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the borrower-facing UI for login, the financial dashboard, settlement requests, and AI negotiation letter generation.	React.js, Vite, Axios, Recharts
API Gateway	Routes and validates all client requests to the appropriate backend endpoints, handling CORS, request parsing, and error responses.	FastAPI (Uvicorn), Nginx reverse proxy

Microservices	Executes core business logic: authentication and authorization, financial/settlement calculations, and AI-driven negotiation letter generation with fallback logic.	FastAPI routers, python-jose (JWT), Google Gemini API, rule-based fallback engine
Database	Persists user accounts, loan records, settlement data, and AI negotiation history for retrieval by the application layer.	SQLite, SQLAlchemy ORM